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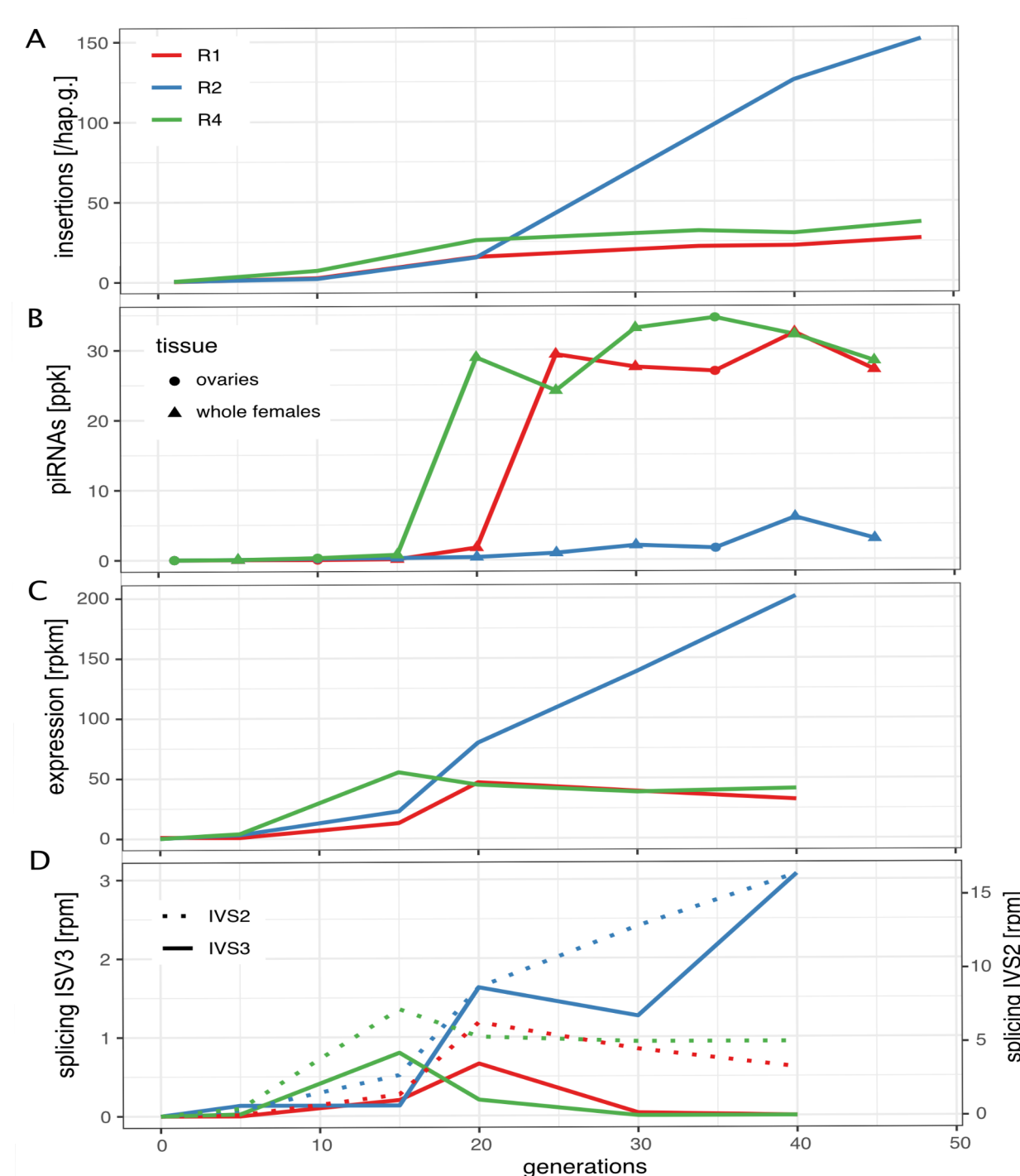


Abstract

To study how silencing against a newly invading transposable element is established, we introduced the P-element, one of the most widely studied eukaryotic transposons, into the naive lines of *Drosophila erecta*. We monitored the P-element invasion for 50 generations in 3 replicates by sequencing the genomic DNA, the small RNAs and the transcriptome at regular intervals. In two out of three replicates, the spread of the P-element was curbed by the piRNA pathway within 20 generations. siRNAs against the P-element were observed before piRNAs suggesting that siRNAs may be responsible for establishing the initial response against a newly invading TE. The emergence of piRNAs did not result in lower P-element expression levels but prevented the splicing of a P-element intron. Interestingly the spread of the P-element could not be stopped in one replicate, despite the presence of P-element insertions in piRNA clusters and siRNAs against the P-element. Although the piRNA pathway was functional in this replicate, the ping-pong cycle failed to respond to the P-element invasion, resulting in the absence of sense piRNAs that bind to Ago3. We investigate why the P-element could not be silenced in this replicate and discuss the impact of this observation on our understanding of the silencing of TE invasions.

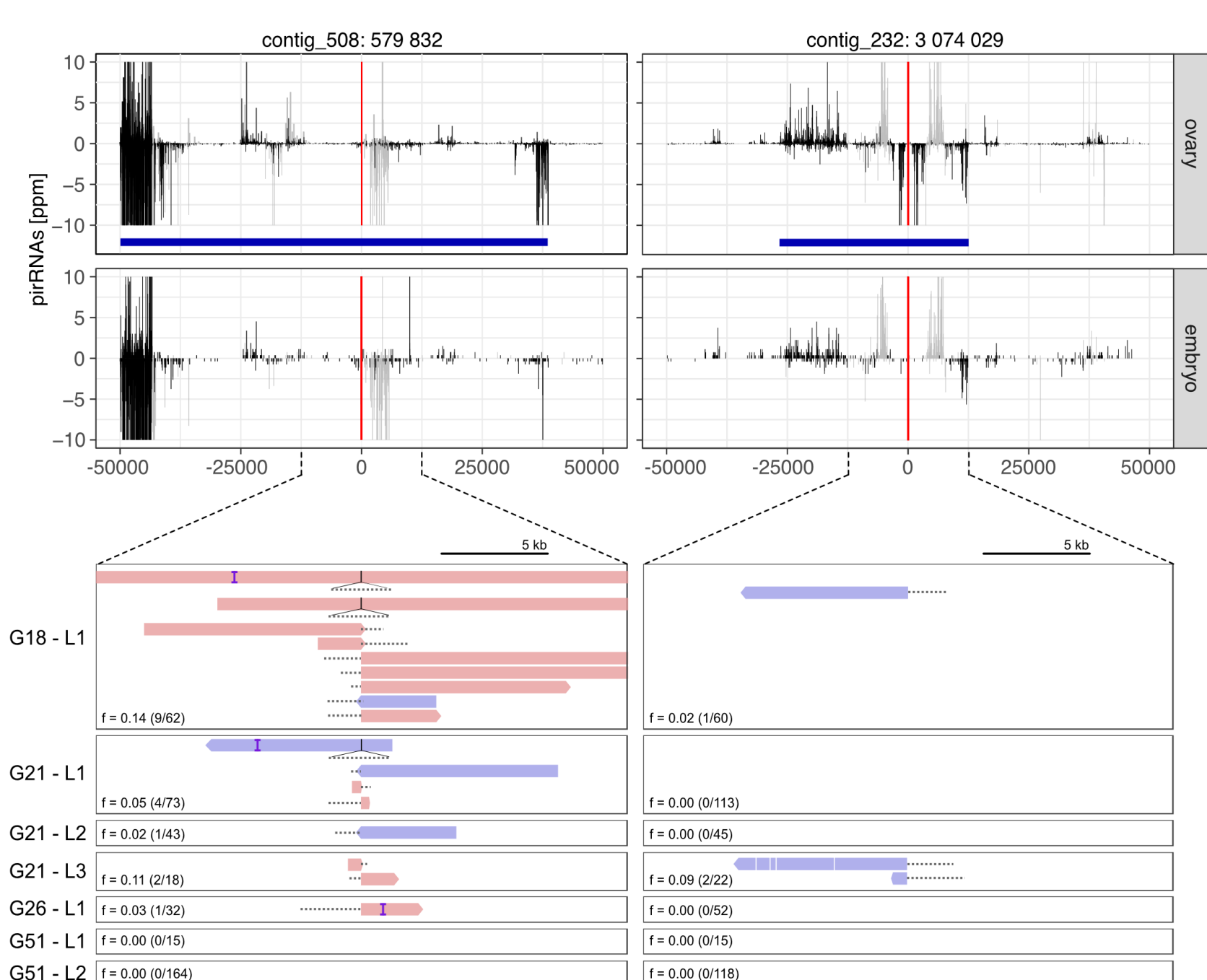
Results

1. P-element invades *D. erecta* populations



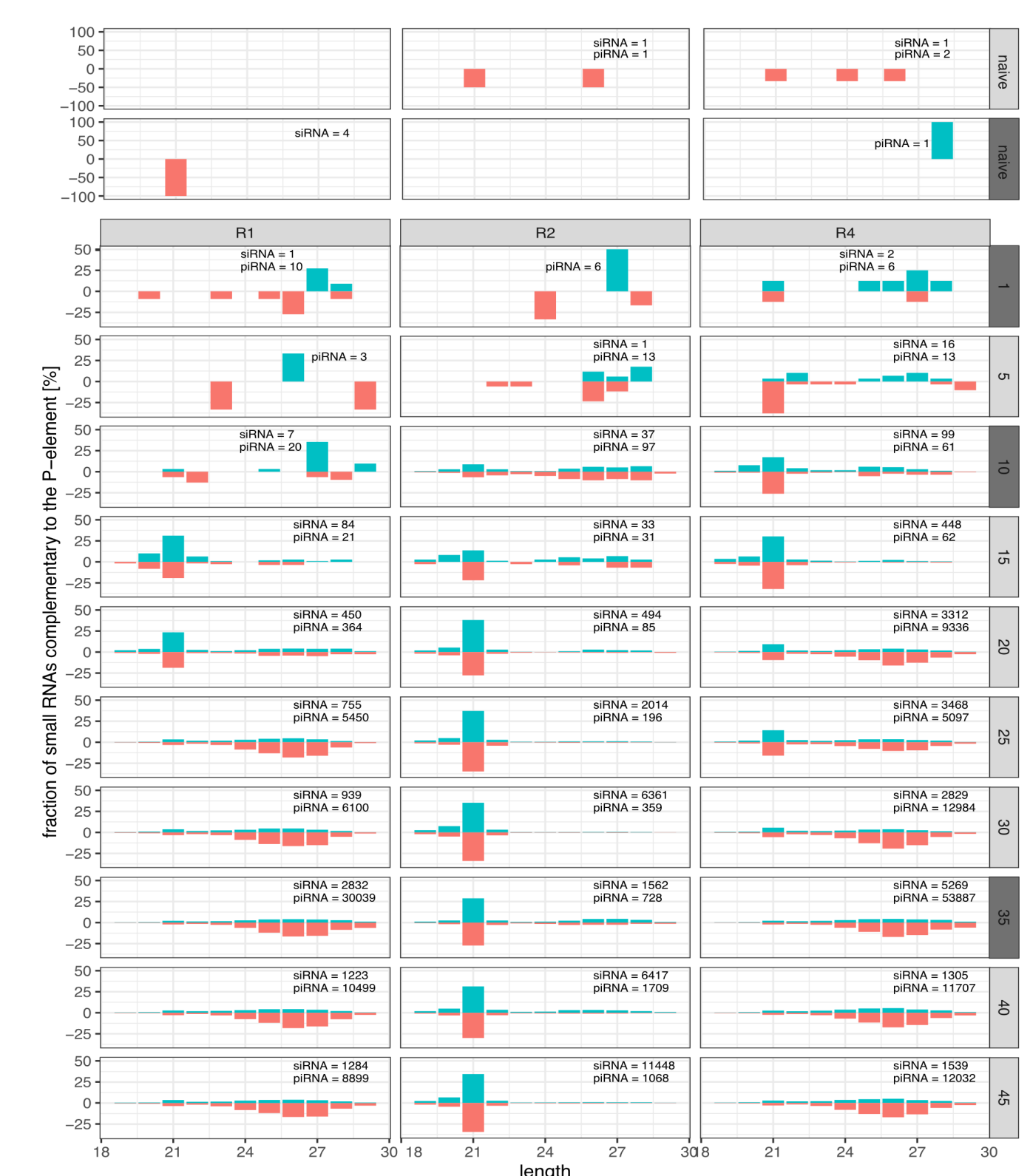
- P-element rapidly increases in copies.
- R2 accumulates a staggering 150 copies by G50.
- piRNAs emerge after generation 20 except R2.
- RNA expression is not controlled during the invasion.
- Splicing of P-element intron IVS3 is regulated in germline to control P-element.

2. P-element insertions in piRNA clusters in R2



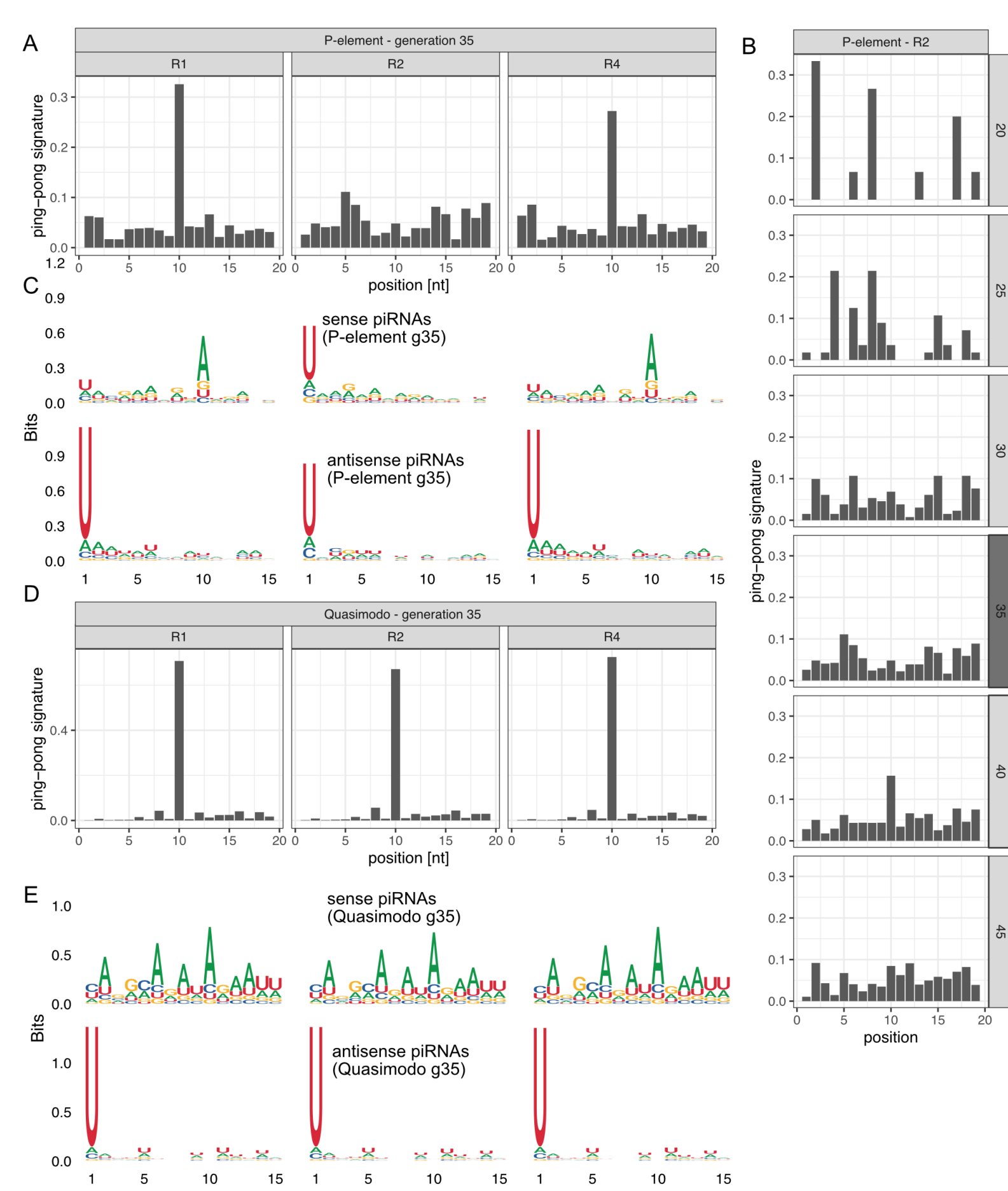
- P-element is inserted in two actively transcribing piRNA clusters in R2.
- Cluster insertions are supported by long reads in many generations.

3. siRNAs emerge before piRNAs during the invasion



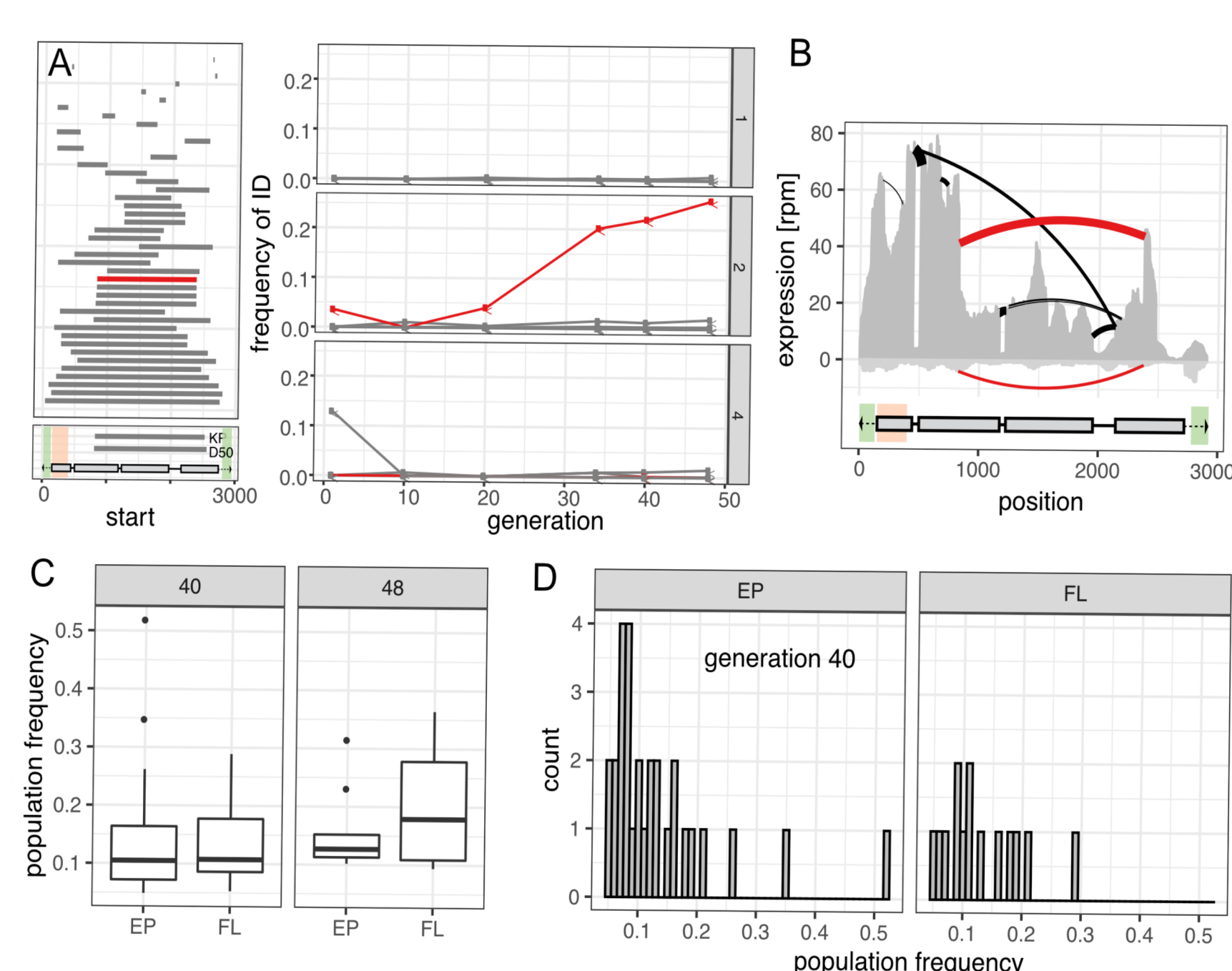
- siRNAs are produced first during the P-element invasion.
- piRNAs emerge after G20 for replicates R1 and R4.
- R2 fails to produce enough piRNAs against P-element.

4. R2 doesn't show ping-pong signature for P-element



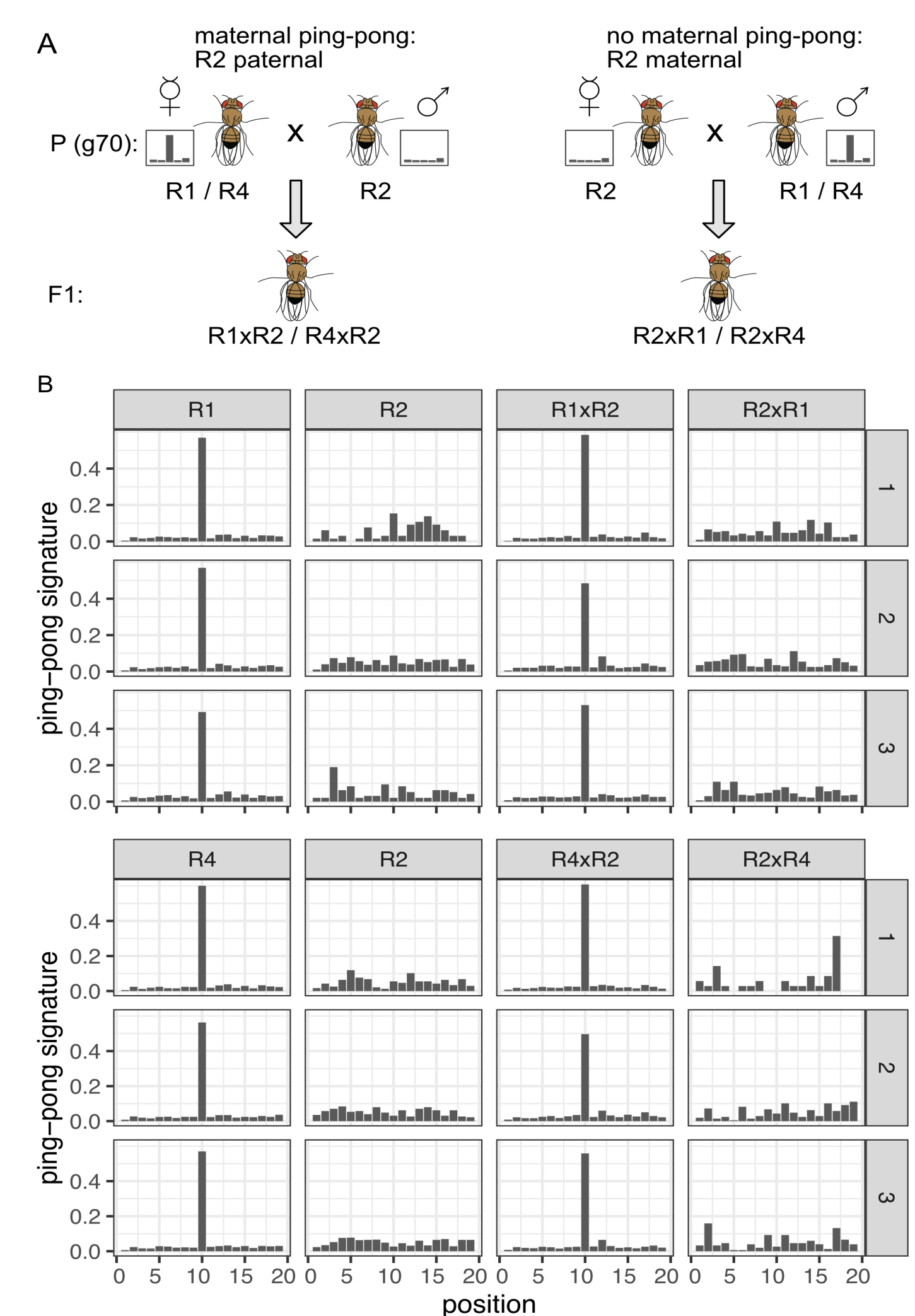
- Replicate 2 fails to produce ping-pong signature.
- We confirm ping-pong signatures in other replicates R1 and R4 with a strong signal.
- It is further confirmed by the enrichment of A bias at the 10th nucleotide.
- Quasimodo in R2 produces ping-pong signature which concludes ping-pong machinery is functional in R2.

5. An internal deletion in R2 increases in copies



- In replicate 2, an internal deletion (EP) increases in copies in comparison to other internal deletions.
- EP's size resembles close to KP like P-element in *Drosophila melanogaster*.
- It has expression levels close to full length.
- Although R2 has high copies of EP element, it's not controlling full length from increasing in copies.
- The population frequency of EP is almost same like full length and it's not positively selected.

6. No ping-pong after G70 in R2 and maternal piRNAs in R2 are defective



- R2 doesn't produce ping-pong signal until G70.
- Maternal piRNAs in R1 and R4 trigger ping-pong in R2 background.
- Reciprocal crosses show that the genome has little effect on establishing ping-pong.

References

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Acknowledgments

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FWF

Der Wissenschaftsfonds.